



SUBMISSION DOCUMENT 06: IMPLEMENTATION & FEASIBILITY PLAN

Phased Deployment Roadmap, Risk Management & Institutional Rollout Strategy

Category: Operational Feasibility, Deployment Timeline & Change Management
Evaluation Weightage Alignment: 15% (Technical Feasibility) + 10% (Scalability & Sustainability)

1. 4-Phase Institutional Implementation Roadmap

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title Digital Campus Institutional Rollout (12-Week Roadmap)	
dateFormat YYYY-MM-DD	
section Phase 1: Setup & Master Ingestion	
Cloud Firestore Schema & Security Rules	:2026-10-01, 14d
Student & Faculty Master Data Ingestion	:2026-10-08, 14d
section Phase 2: Core ERP & Attendance	
Smart Attendance Radar & Biometric Pilot	:2026-10-22, 14d
Timetable, Fee Portal & Digital Bonafides	:2026-10-29, 14d
section Phase 3: AI Engine Activation	
Voice Assistant & Speech NLU Deployment	:2026-11-12, 14d
Predictive Performance & Dropout EWS Radar	:2026-11-19, 14d
section Phase 4: Institutional Scaling	
Parent Portal & Transit Bus GPS Integration	:2026-12-03, 14d
Campus-Wide Multi-Tenant Go-Live	:2026-12-17, 7d

Detailed Phase Breakdown

- **Phase 1: Foundation & Data Ingestion (Weeks 1–2)**
 - Setup multi-tenant Cloud Firestore database and security rules.
 - Ingest legacy student roll numbers, enrolled courses, and faculty rosters via automated CSV bulk migration scripts.
 - Issue cryptographic student digital identities.
- **Phase 2: Core ERP & Operational Rollout (Weeks 3–4)**
 - Deploy 1-Tap Attendance Check-In in classrooms; verify geofence beacons in lecture halls.
 - Activate the Dynamic Timetable with faculty substitution notification desks.
 - Launch Instant Bonafide Certificate generation and UPI fee payment channels.
- **Phase 3: Applied AI Activation (Weeks 5–6)**
 - Calibrate the Multi-Variate SGPA Regression model ($R^2=0.91$) on historical department cohort grades.
 - Activate the 4-Pillar Early Warning Dropout Risk engine; train academic counselors on 1-tap automated intervention workflows.
 - Launch the Multi-Modal AI Voice Assistant for instant student self-service.
- **Phase 4: Parent Portal & Full Scale Go-Live (Weeks 7–8)**
 - Onboard guardians to the Parent Portal via secure SMS links.
 - Integrate live bus GPS transit tracking with route driver hotlines.
 - Conduct mock AICTE/NAAC digital accreditation compliance audits.

2. Technical Feasibility & Prerequisites

Requirement Pillar	Institutional Requirement	Digital Campus Implementation	Feasibility Status
Client Devices	Android, iOS, or Web Browser	Pure Flutter responsive web and native mobile binaries	✔ 100% Feasible
Server Infrastructure	Dedicated on-premise servers	Serverless Cloud Firestore + Firebase Authentication	✔ Zero Server Maintenance
Network Resilience	Variable Wi-Fi on campus	Full local memory caching & offline-first operation	✔ Works 100% Offline
Hardware Terminals	Expensive biometric turnstiles	Uses student smartphone GPS + dynamic QR code matching	✔ Zero Hardware Capex

3. Risk Management & Mitigation Strategies

Identified Risk	Potential Impact	Built-in Mitigation Mechanism in Digital Campus
1. Intermittent Campus Wi-Fi	App fails during lecture attendance roll-calls.	Offline-First Reactive Architecture: Attendance is stored locally in client memory and synchronizes automatically upon reconnecting.
2. Proxy Attendance Fraud	Students sharing QR screenshots via WhatsApp.	Dynamic Time-Expiring QR Codes: Attendance QR codes refresh every 15 seconds and enforce a 50-meter GPS geofence boundary.
3. Faculty Hesitation to Adopt	Reluctance to abandon paper register books.	Zero-Friction UI: Marking attendance takes a single tap on the screen; substitute classes can be assigned in < 10\$ seconds.
4. Student Data Privacy	Exposure of academic records and student phone numbers.	Role-Based Access Control (RBAC): Students can only view their own records; student contact numbers are shielded from peer discussion groups.

4. Change Management & Training Plan

- Faculty Workshops (2-Hour Hands-on Session):** Training professors on 1-tap QR attendance, substitute period selection, and interpreting student at-risk flags.
- Student Onboarding Drive:** 10-minute orientation during orientation week showing students how to generate instant Bonafides, ask voice queries, and monitor safe bunks.
- Parent Onboarding SMS:** Automated SMS sent to registered parents with their private login link and assigned faculty mentor contact.